Chenwei (Denny) Gan

Full-stack software engineer
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Summary

Full-stack software engineer working with TypeScript/React, Python/FastAPI, PostgreSQL and AWS. Built and shipped two products end to end: an AWS-hosted web app and an iOS app on the App Store. Master of Information Technology (First Class Honours), University of Auckland, with six years of international business operations experience, which means I ask what a system is for before designing how it works. Post-Study Work Visa, no sponsorship needed.

Technical Skills

Languages
TypeScript, JavaScript, Python, Dart, Swift, SQL, Java
Frontend
React, Vite, Tailwind CSS, React Query, Zustand, Vue.js, Svelte (SvelteKit)
Backend
Node.js, Express, Fastify, FastAPI, SQLModel, Prisma, Zod, REST APIs, WebSocket, OpenAI API
Databases
PostgreSQL, MongoDB (Mongoose), SQLite
Cloud & DevOps
AWS (EC2, RDS, S3, CloudFront, Cognito, Lambda), Terraform, GitHub Actions, Docker, Git
Mobile
Flutter, Riverpod, native Swift (PhotoKit, AVFoundation)
Security
Authentication and authorization (RBAC/ABAC), OAuth, Cognito

Projects

echoType (Web) echotype.ink May 2026 - Present
Solo-built low-pressure typing web app for internalising texts you choose (poems, speeches, quotes) by typing them on a loop, with your own notes shown above each phrase; live since July 2026.

- Anchored notes exactly above their phrase by measuring each character's pixel position in a hidden mirror element, re-measuring only on layout change; a stored snapshot check flags notes that no longer match after the text is edited.
- Defined the AWS infrastructure in Terraform (VPC, EC2, RDS, S3, CloudFront, Cognito, ACM, IAM, SSM, Lambda) and implemented Cognito email/password and Google sign-in with account linking.
- Set up one-click GitHub Actions deploys using OIDC and SSM Run Command, with no inbound SSH: the API runs as a Docker container on EC2 and applies Prisma migrations on startup, and the SPA ships to S3 behind CloudFront.

WakeTok (Mobile) apps.apple.com/nz/app/waketok/id6797156391 Jul 2026 - Present
Vertical-scrolling mobile app for rediscovering your own photo and video library; designed, built and shipped solo. Free on the App Store (iOS 18+) since October 2026, with no account, ads, analytics or uploads.

- Flutter UI over a native Swift layer for PhotoKit library access, AVFoundation video playback and Live Photos, connected through Pigeon typed contracts.
- Browses full libraries including iCloud-only items, tested on a real 40,000-item library; saved browsing selections persist locally in Drift/SQLite.

- Product rules (selection, ordering) live in Dart, with layer boundaries enforced by automated architecture tests, keeping the core reusable for a planned Android port.

Professional Experience

Full Stack Developer (Internship) Jul 2025 - Oct 2025

PILOT (Vanta Capital Trading Ltd) | Auckland

Internship at an Auckland fintech startup, building the core infrastructure of a real-time, risk-guardrailed copy-trading MVP, enabling retail users to replicate trades under strict access control and audit requirements.

- Designed and implemented a hybrid RBAC–ABAC authorization model in FastAPI with a three-layer structure spanning user claims, RBAC, and ABAC policy evaluation.
- Built the group-based membership system that governs all interactions between professional traders and retail followers, including following, joining, and trade signal distribution, ensuring every membership action is permission-checked and audit-logged before sensitive trade data reaches any user.
- Independently developed the Trade Order Card system in React and TypeScript with Zustand, creating a single base component with four functional variants and a separate audit layer.
- Implemented a complete RTMP-to-HLS live streaming pipeline (REST + WebSocket + webhook-based session management) so professional traders can broadcast their screens to followers.

GTM (Go-to-Market) Supervisor Nov 2021 - Mar 2023

Tineco

Led go-to-market execution across Europe and APAC, while serving as the business-side system owner of multiple enterprise operational system rollouts.

- Business-side system owner for the GTM module of the Global Order Management System (GOMS), partnering with IT on requirements gathering, user acceptance testing (UAT), and rollout across European and US subsidiaries.
- Co-defined the functional design of a Java-based internal promotion management system within the GOMS operational ecosystem, replacing an external SaaS tool and supporting multi-channel campaign coordination, conflict detection, and real-time integration with sales and finance workflows.

Event Planning Manager May 2021 - Nov 2021

SUNRA

Led nationwide retail campaigns, while serving as the business-side lead on campaign-related IT systems.

- Defined requirements and worked with internal and external teams to develop a nationwide online lottery system, enabling retail staff to register purchases via receipt and e-bike ID for real-time prize draws.
- Defined functional requirements and conducted UAT to support campaign-side IT readiness for nationwide rollout.

Country Retail Manager Jul 2017 - May 2021

Transsion

Led Transsion's TECNO retail operations in Tanzania (300+ staff, 1,000+ stores), while contributing business-side input to internal retail and channel IT systems.

- Worked with the IT and data analytics teams on the retail and channel management systems (app and web) and sell-out/KPI dashboards, contributing UX feedback and data structure input from a frontline operations perspective.

Education

University of Auckland 2025

Master of Information Technology

First Class Honours, GPA 8.1/9. Includes an integral industry internship at Pilot.

Jiangxi University of Finance and Economics 2017

Bachelor of Economics, Finance (International Finance)